Lecture 31.10: Hierarchical Clustering — Agglomerative vs Divisive











📌 Topic Overview

Comprehensive conceptual analysis of Hierarchical Clustering. Contrasts bottom-up Agglomerative clustering against top-down Divisive clustering, and provides mathematical derivations for Single, Complete, Average, and Ward linkage criteria.

📋 Key Learning Objectives

- Concept of hierarchical tree-based clustering without pre-specifying K.
- Agglomerative Clustering (Bottom-Up): Iteratively merging the two closest clusters.
- Divisive Clustering (Top-Down): Iteratively splitting one master cluster into subclusters.
- Linkage Criteria: Single (min), Complete (max), Average (mean), and Ward (variance minimization).
- Trade-offs: Sensitivity to outliers and computational complexity O(N^2 log N).

🔍 In-Depth Conceptual Breakdown

- **Hierarchical Clustering Ka Core Idea**: K-Means mein hume shuruat mein hi batana padta tha ki $K = 3$ ya $K = 4$. Lekin **Hierarchical Clustering** mein hume pehle se K decide nahi karna padta! Yeh poore data ka ek family tree (hierarchy) bana deti hai.

- **Agglomerative (Bottom-Up) Approach (Sabse Popular)**:

- 1. Shuruat mein har ek data point apna khud ka individual cluster hota hai (N points = N clusters).

- 2. Sabse paas wale do clusters ko aapas mein merge kar diya jata hai ($N-1$ clusters bache).
- 3. Yeh merging tab tak chalti rehti hai jab tak end mein ek single giant cluster na ban jaye!

- **Divisive (Top-Down) Approach**: Iska ulta hota hai. Saare points 1 cluster se start hote hain aur recursive splits karke single points tak pahunchte hain.

- **Do Clusters Ke Beech Distance Kaise Nikalein? (Linkage Criteria)**:

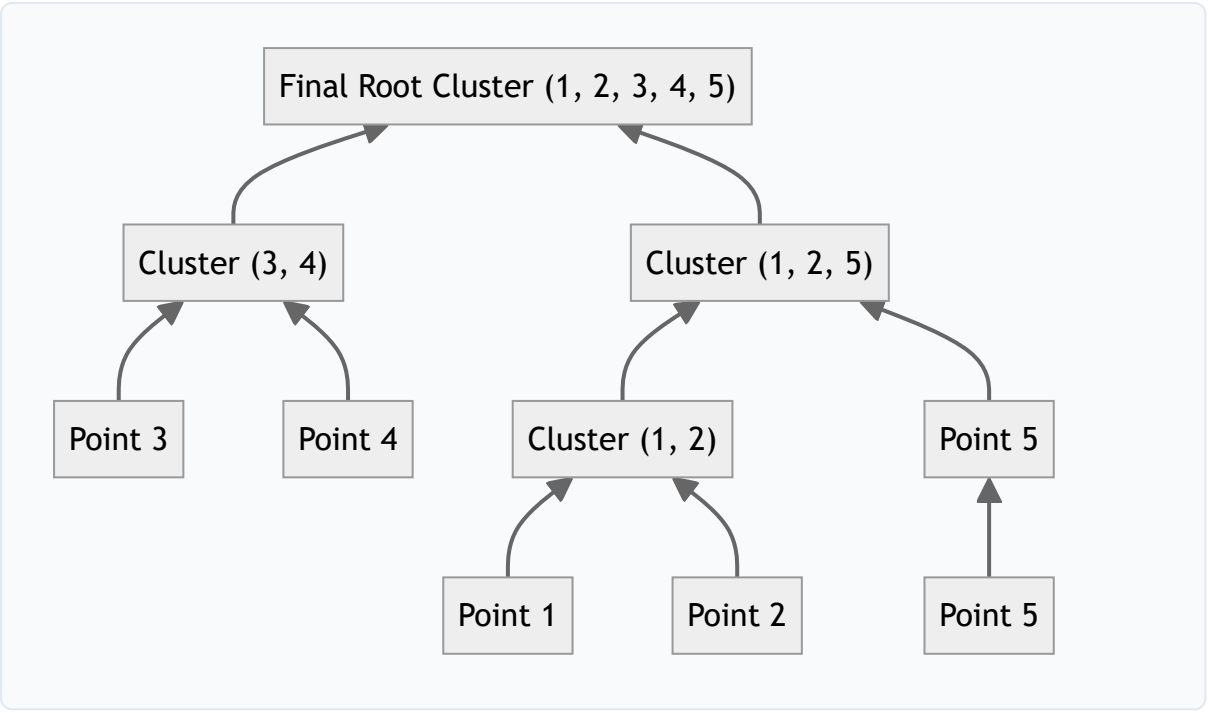
- **Single Linkage**: Cluster A aur Cluster B ke do sabse paas wale points ka distance: $d_{min}(A,B) = \min d(x,y)$. (Chaining effect ka khatra).
- **Complete Linkage**: Do sabse door wale points ka distance: $d_{max}(A,B) = \max d(x,y)$. (Compact, spherical clusters banata hai).
- **Average Linkage**: Dono clusters ke saare pairs ka average distance.
- **Ward Linkage (Default in Scikit-Learn)**: Do clusters ko merge karne par variance (sum of squared errors) mein kitna izafa hoga: ΔVar. Jo merge variance ko sabse kam badhaye, usi ko merge kiya jata hai!

- **Pure Theory Lecture**: Is lecture mein ma'am ne blackboard par Agglomerative vs Divisive aur Linkage formulas explain kiye hain, koi code nahi hai.

📋 Linkage Criteria Comparison Matrix

Linkage Method	Mathematical Definition	Cluster Shape Tendency	Outlier Sensitivity
Single Linkage	$d(A,B) = \min_{x \in A, y \in B} d(x,y)$	Can follow elongated shapes	High (prone to chaining noise)
Complete Linkage	$d(A,B) = \max_{x \in A, y \in B} d(x,y)$	Compact, equal diameter spheres	Low (sensitive to outer boundaries)
Average Linkage	$d(A,B) = \frac{1}{\|A\|\|B\|} \sum d(x,y)$	Balanced variance clusters	Moderate
Ward's Linkage	$\Delta \text{ESS} = \frac{\|A\|\|B\|}{\|A\|+\|B\|} \|\mu_A - \mu_B\|^2$	Compact, spherical, variance-optimal	**Lowest (Industry standard default)**

📊 Agglomerative (Bottom-Up) Merging Flow



🔢 Mathematical Formulation

Ward's minimum variance criterion for merging clusters A and B:

$$\Delta I_{AB} = \frac{N_A N_B}{N_A + N_B} \|\mathbf{m}_A - \mathbf{m}_B\|^2$$

where $\mathbf{m}_A$ and $\mathbf{m}_B$ are the centroid mean vectors of clusters A and B.

📌 Note on Code

This is a pure conceptual, mathematical, and blackboard theory lecture. In accordance with course guidelines, no synthetic code is appended.